

CRM Track C Prompt Logbook

Mandi @ 36: AI-Driven Sentiment Analysis of Customer Feedback

Prepared by: Malla Venkata Sai Ashish | Roll Number: B31-25

Purpose. This logbook records the practical prompts used or reconstructed from the Mandi @ 36 workflow. AI assisted with code scaffolding, interpretation, dashboard communication, and writing. It did not create raw customer reviews or replace deterministic analytics.

Prompt Record

The prompts below are written in a reproducible form so that a reviewer can understand what information was shared, what output was requested, and how the result was checked. Tools such as Copilot, Claude, Codex, and ChatGPT were used selectively according to the task. Any generated code or narrative was reviewed against the local files and project outputs before inclusion.

No.	Tool and task	Prompt text	Output, project use, and reflection
1	Codex Schema inspection	I have Apify Google Maps scraper JSON exports. Inspect the actual schema without changing any raw file. Identify the review-list field and map location, review text, rating, date, reviewer name, and source URL. Report missing or ambiguous fields before proposing code.	Produced a field-mapping checklist and schema-validation approach. Used to keep raw files intact and document the available evidence.
2	GitHub Copilot JSON preparation	Convert the Google Maps scraper JSON into a usable analytical table in Python. Write a defensive loader that accepts nested review arrays, keeps branch name and source URL, normalizes rating and date, preserves raw text, and writes a separate cleaned output. Do not invent missing values.	Provided a code scaffold for normalized review records. The implementation was adapted after checking the real export structure and saved outputs separately from raw data.

No.	Tool and task	Prompt text	Output, project use, and reflection
3	Claude Text cleaning	Suggest a privacy-minimizing text-cleaning sequence for public restaurant reviews. Preserve the original review text, create a clean analysis field, and list limitations for emojis, mixed languages, sarcasm, and missing text.	Produced cleaning steps and limitations. Used to explain preparation choices and the requirement for human interpretation.
4	Codex Sentiment baseline	Create a transparent sentiment-analysis baseline for restaurant review text using interpretable positive and negative lexicons. Return a label and score, keep the logic separate from any LLM call, and state why it must not be presented as CSAT or NPS.	Generated a deterministic baseline design. It informed the derived sentiment labels and the report's caution that model labels are screening signals.
5	ChatGPT Topic and aspect interpretation	Explain, in plain language, how keyword aspects and TF-IDF plus NMF topic modeling can identify recurring restaurant experience areas such as food, service, ambience, waiting time, and value. Include limits and manager-validation steps.	Created a plain-language explanation. Used to make the methodology and finding sections understandable without overstating causation.
6	GitHub Copilot Dashboard design	For a Streamlit restaurant Voice of Customer dashboard, propose 5 to 7 visuals that help a branch manager move from review text to action. Include sentiment distribution, branch comparison, word cloud, trend, complaint themes, anomaly flags, and action cases.	Suggested dashboard sections and visual roles. The final application shows selected visuals and preserves the distinction between collected, derived, and predicted evidence.
7	Project AI Analyst CRM action query	What actions should we take for Banjara Hills? Use only the supplied compact JSON evidence for branch summary, recurring aspects, topics, alerts, and CRM cases. State evidence, confidence, recommended owner, next action, and what a manager should validate.	Returned an evidence-grounded action narrative. A full page screenshot and example output are included in the report; the manager remains the final decision-maker.
8	Claude Ethics review	Review this public-review sentiment workflow for privacy, bias, transparency, and over-automation risks. Give practical controls for a restaurant CRM process, including consent, retention, human review, and treatment of sarcasm or language variation.	Produced a risk-and-control checklist. Used to strengthen the ethical, legal, and bias considerations section.
9	ChatGPT CRM roadmap	Propose a 12-week phased implementation roadmap for a restaurant feedback-to-action workflow. Include pilot, ownership, response SLA, staff training, monitoring, calibration, risks, and mitigation. Mark recommendations clearly rather than presenting them as completed operations.	Provided a planning structure. It was converted into the Gantt chart, CRM scorecard, and risk table after adapting it to the project scope.

No.	Tool and task	Prompt text	Output, project use, and reflection
10	Codex Report QA	Check the report against this brief: 10 to 12 pages, Times New Roman 12 pt body text, 1.5 spacing, 1-inch margins, justified paragraphs, APA references, clickable deployment and repository links, Google Maps sources, visuals, and no unsupported claims. Identify only evidence-backed corrections.	Produced a quality-assurance checklist. The final report was rendered and visually checked, with figures, links, citations, and claims reviewed before submission.

Verification and Reflection

All computed numerical results in the CRM report are sourced from the local pipeline artifacts, including dataset summary, branch summary, NLP outputs, anomaly outputs, topic model outputs, and CRM cases. AI assistance was used to draft explanations and structure, but it was not used to manufacture review records, metrics, or customer outcomes. The strongest practice is to retain evidence links, label derived results, and require human validation before customer-facing action.

Technical Evidence Used

Pipeline. `src/pipeline.py` reads raw exports, normalizes data, runs analytics, and writes result artifacts without modifying raw review files.

Sentiment and aspects. `src/analytics/nlp.py` implements a transparent lexicon baseline and keyword aspect extraction, each marked derived and unvalidated.

CRM cases. `src/analytics/crm.py` ranks recurring experience-area signals and attaches a suggested owner, action, status, and review cycle.

AI routing. `src/agents/orchestrator.py` detects branch, comparison, period, and action intent before passing compact evidence to the CRM recommendation agent.